

Digital Time Machine: A Virtual Reality Reconstruction of the Southwestern Zeitgeist Through the Lens of Clark Hulings' Artistic Legacy

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Figure 1: "The Woodbearers of Chimayó" - Clark Huling's painting in Virtual Reality.

ABSTRACT

This research demo showcases the "Digital Time Machine," a virtual reality (VR) experience designed to transport users to 1974 Chimayó, New Mexico, as captured through the artistry of Clark Hulings. By integrating photorealistic VR environments, artistic stylization, and user-centered design, the project demonstrates the potential of immersive technology to deepen cultural appreciation, historical understanding, and artistic engagement. Developed in collaboration with the New Mexico Museum of Art Vadem Contemporary, this project exemplifies how VR can bridge temporal and spatial divides, allowing museum audiences to step into the world of Hulings' iconic painting.

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Index Terms: Virtual reality, Clark Hulings, cultural preservation, photorealistic environments, art education, museum exhibits, Unreal Engine.

1 INTRODUCTION

Working for the New Mexico Museum of Art Vadem Contemporary, our team was tasked with designing a Virtual Reality (VR) experience for Clark Hulings' 1974 painting titled "The Woodbearers of Chimayó." The museum reached out to us with the vision of creating an immersive application that would allow users to step into the painting's world and explore its historical and cultural context. This collaboration aimed to bring Hulings' work to life in a way that bridges the gap between traditional art appreciation and cutting-edge technology.

2 MOTIVATION AND OBJECTIVES

The project addresses the challenge of making static art more interactive and engaging for contemporary audiences. Hulings' 1974 painting of Chimayó provides a rich narrative and aesthetic lens through which to explore the cultural and historical nuances of the American Southwest. The Santuario de Chimayó, a Catholic shrine built by Don Bernardo Abeyta in the late 19th century, is the most iconic and central part of Chimayó's culture [1]. The shrine's ori-



Figure 2: Santuario de Chimayó [2].

gins trace back to Abeyta's discovery of a crucifix buried in the dirt at the site, which was believed to possess miraculous healing properties—a belief that endures to this day. Annually, over 300,000 pilgrims visit the Santuario de Chimayó, with Holy Week being its busiest period [3].

The objectives of the “Digital Time Machine” included:

- **Recreating the Chimayó landscape** with historical accuracy and artistic fidelity.
- Providing an intuitive VR experience tailored for museum-goers, including those with limited technical familiarity.
- Enhancing understanding of the historical and artistic contexts surrounding Hulings' work.

3 TECHNICAL APPROACH

The development leveraged Unreal Engine 5.4, utilizing its Nanite and Lumen systems for high-fidelity rendering and dynamic lighting. The terrain was modeled using LiDAR data from the United States Geological Survey (USGS), ensuring geographical precision. Key elements of the project included:

- **Artistic Stylization:** A post-processing pipeline combined photorealistic textures with painterly effects, maintaining fidelity to Hulings' style while immersing users in the 3D environment.
- **Interactive Elements:** Users can explore the recreated town, encountering animations such as burros and other dynamic elements that bring the painting to life.
- **Accessibility and Usability:** Designed for first-time VR users, the system employs seated navigation, simple controls, and a self-resetting application for ease of use.

4 USER STUDIES AND FEEDBACK

Initial user studies were conducted to evaluate motion sickness, accessibility, and engagement. Participants reported increased appreciation for Hulings' painting after experiencing the VR environment, noting a deeper understanding of the 3D spatial relationships within the scene, enhanced recognition of cultural landmarks, such as the Santuario de Chimayó, and an emotional resonance with the historical and artistic narrative.

5 APPLICATIONS AND IMPACT

The “Digital Time Machine” is intended as a permanent exhibit at the New Mexico Museum of Art Vladem Contemporary. It serves as a case study in the use of VR for:



Figure 3: The path the burros take across the painting, with the start positions of the burros highlighted in green.

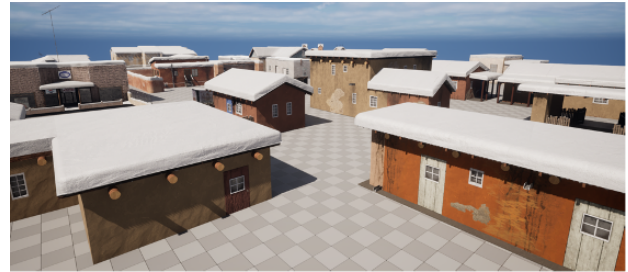


Figure 4: Unique buildings in Unreal Engine.

- **Art Education:** Providing immersive, interactive ways to engage with historical art.
- **Cultural Preservation:** Documenting and recreating historical sites for future generations.
- **Museum Innovation:** Setting a precedent for VR integration in traditional exhibit spaces.

6 CONCLUSION AND FUTURE WORK

This research demo highlights the transformative potential of VR in redefining how audiences experience art and history. Future directions include expanding the “Digital Time Machine” framework to other artists and locales, refining accessibility features, and exploring haptic feedback for deeper immersion.

By blending technological innovation with artistic reverence, this project not only reconstructs the past but also reimagines how it can be experienced, offering an unparalleled bridge between art, history, and technology.

SUPPLEMENTAL MATERIALS

More about this project can be viewed at <https://www.vastlab.com/projects/woodbearers> released under a CC BY 4.0 license.

ACKNOWLEDGMENTS

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